

SUNSTONE

The purpose of this lab sheet is to familiarize yourselves with the Sunstone web interface of Opennebula by building your first Cloud VM. This will be the main interface you will use throughout the course so you must make sure you become confident in its use. Our endgame for this lab sheet is for you to create your own first customized virtual machine, the first node in your Cloud. In order to do this we need to:

1. Configure a Network
2. Configure a Host
3. Configure a Datastore
4. Create an Image
5. Create a Template
6. Create a Virtual machine and build it

CREATE THE HOST BRIDGE

Warning

Backup all the files that are modified in this section and move them to your home folder before making changes to them.

You will need to have your main interface connected to a bridge. We will do the following example with `ens33` but the name of the interface may vary (however, this should be `ens33` on your computers). An OpenNebula requirement is that the name of the bridge should be the same in all nodes.

To do so, type **gedit** `/etc/sysconfig/network-scripts/ifcfg-ens33`

```
DEVICE=ens33
BOOTPROTO=none
NM_CONTROLLED=no
ONBOOT=yes
TYPE=Ethernet
BRIDGE=br0
```

And add a new `br0` file with **gedit** `/etc/sysconfig/network-scripts/ifcfg-br0`

```
DEVICE=br0
TYPE=Bridge
IPADDR=148.197.28.??? (Replace these questionmarks with your Prometheus IP address) PREFIX=23
GATEWAY=148.197.28.254
DNS1=148.197.28.251
DOMAIN=netlab.ee.port.ac.uk
ONBOOT=yes
BOOTPROTO=static
NM_CONTROLLED=no
```

After these changes, restart the network:

```
systemctl restart network.service
```

Once this process is complete reboot your Host and check your Internet connectivity. If everything still works ok then go into OpenNebula and complete the following tasks.

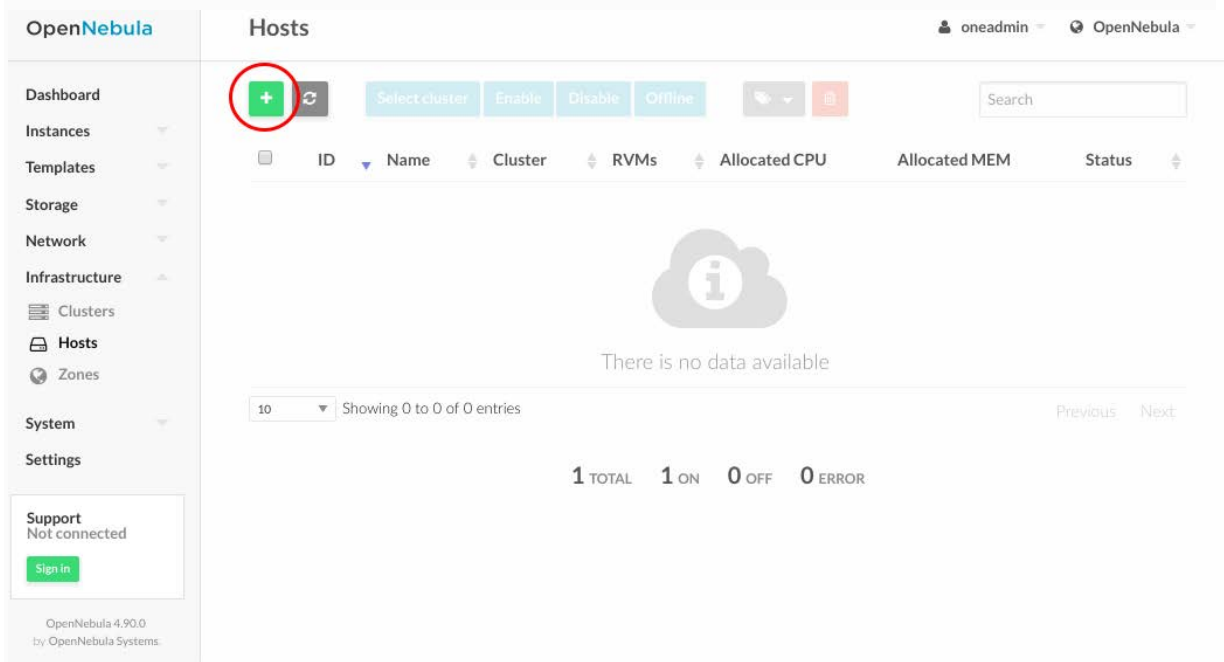
ADDING A HOST

In this step we will register the node we have installed in the OpenNebula Front-end, so OpenNebula can launch VMs in it. This step can be done in the CLI or in Sunstone, the graphical user interface. Make sure sunstone is running and then configure the services to run automatically at boot time using:

```
service opennebula restart; chkconfig opennebula on
service opennebula-sunstone restart; chkconfig opennebula-sunstone on
```

Adding a Host through Sunstone

Open the Sunstone website. In the left side menu go to Infrastructure -> Hosts. Click on the  button.



Then fill-in the FQDN of Prometheus in the **Hostname** field. And click **Create**. Finally, return to the **Hosts** list, and check that the Host switch to , status. It might take somewhere between 20s to 1m to initialize. Try clicking on the refresh button to check the status more frequently. If the host turns to  state instead of , check the `/var/log/one/oned.log`. Chances are it's a problem with your Hostname or your passwordless ssh!

ADDING A DATASTORE

Click on *Storage -> Datastores*. You should have 3 preconfigured Datastores already. Check that they are there and do not alter them.

ADDING NETWORKS

In the **Networks tab** add a new network with a name of **Bridge** then go to **conf** and use bridge **br0** and choose your physical device as **ens33**. Then go to **addresses** and set your **first IP address** as **148.197.28."Prometheus+50"** . Choose size **1**. Go to the **Context Tab** and set the Network address to **148.197.28.0**, the Netmask to **255.255.254.0**, the Gateway to **148.197.28.254** and the DNS to **148.197.28.251**. Click **Create**.

ADDING IMAGES

We will need 2 Images at this point. One is the image that we will use as a DVD to build the operating system and the other is the image that we will use as hard drive space for our Virtual machine. Go to *Storage -> Images*.

Image 1: Add another image, call it **vPrometheus0**, change the *type* to **Generic Storage Datablock** and select **YES** to the **persistent** option. Choose **Empty Disk Image** and give it a size of **10 GB**, go to the *Advanced Options* and select *Format* **qcow2** and click **create**. This should be ready immediately.

Image 2: Add an Image, name it "**Centos**" and set the type to **Readonly CD-ROM**. In Image Location select **Upload** and click on *Choose file* and select the CentOS iso from your home directory. Click on **create** and you will notice that the image is now there but **LOCKED** and it will remain so until the iso is uploaded from your home folder. Refresh every once in a while until it turns to **READY**.

ADDING A TEMPLATE

Now that we have our images we need to create a template for our VM. Go to *Templates -> VMs* and click on add. Name it **vP0**, choose **KVM** for the *Hypervisor*, give it **2GB** of RAM, Say **Yes** to *Enable Host Resize*, give it **1 physical and 1 virtual** CPU and enable Hot Resize. Then move onto the *Storage tab* and select the **vPrometheus** for **Disk 0** then add **Disk 1** and assign **CentOS** to it. Go to the *Network Tab* and select the **Bridge**. Go to *OS & CPU* and select **x86_64** for the *Architecture*, Choose **disk0** for the **1st Boot** and **disk1** for the **2nd BOOT**. Go to *Input/Output* and select **VNC**. Click **Create** and your template should appear.

ADDING A VIRTUAL MACHINE

Now that we have all the ingredients for our recipe it's time to cook!

Go to *Instances -> VMs* and click add. Select the **vP0** template and name the Vm **vP0**. Check that **capacity**, **disks** and **network** are correct as per your image settings and click **create**. The VM will be created, it will obtain your first useable IP address and its status will be **PROLOG** for a while then if you refresh it it will change to **Booting** and then finally **Running**!

You can now click on the **VNC icon** and connect to your running VM. Go through the installation steps choosing to install a **Gnome Desktop** setting the hard drive partitioning to automatic and network as they should be and wait for it to finish installing. Reboot and you have your first own computer in your Cloud! Congratulations!